

Fencing Hallucination: An Interactive Installation for Fencing with AI and Synthesizing Chronophotographs

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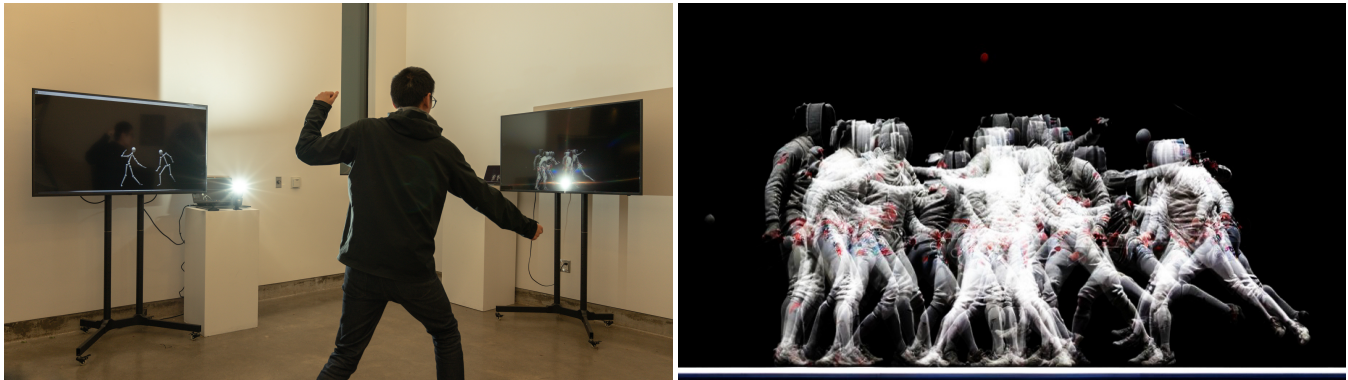


Figure 1: Setup illustration(left) and final result(right). The left image shows the actual setup in a room, and the right image shows the enlarged view of a final chronophotograph result.

ABSTRACT

Fencing Hallucination is a multi-screen interactive installation that enables real-time human-AI interaction in the form of a Fencing game and generates a chronophotograph based on the audience's movement. It mitigates the conflicts between interactivity, modality variety, and computational limitation in creative AI tools. *Fencing Hallucination* captures the audience's pose data as an input to the Multilayer Perceptron (MLP), which generates the virtual AI Fencer's pose data. It also uses the audience's pose to synthesize the chronophotograph. The system first represents pose data as stick figures. Then it uses a diffusion model to perform image-to-image translations, converting the stick figures into a series of realistic fencing images. Finally, it combines all images with an additive effect into one image as the result. This multi-step process overcomes the challenge of preserving both the overall motion patterns and fine details when synthesizing a chronophotograph.

CCS CONCEPTS

• Human-centered computing → Gestural input; • Computing methodologies → Image representations.

KEYWORDS

Image-to-image Translation, Diffusion Model, Human-AI Interaction, Chronophotography, Motion Synthesis

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1 INTRODUCTION

Text-to-image diffusion models such as *DALL-E 2* [OpenAI 2022] became phenomenally popular almost overnight, bringing AI creativity accessible to the general public besides computer scientists and creative technologists. Everyone can use these tools to synthesize a realistic image by providing a text description. The powerful generative ability of these tools makes people presume the authenticity of AI-generated images and, as a result, seek more uniqueness and connectedness in them. The users of text-to-image models may spend hundreds of iterations tweaking their text prompts to generate an ideal image. The iterations are slow, preventing the audience from having a responsive experience using these tools. Text is also limited for describing morphological information. Is there an alternative way to interact with these AI Image Generators with additional modalities and get some instant responses? How can these additional input modalities to the AI system consolidate the relationship between the AI-generated artifacts and the user? How to push the boundary in the complexity of AI-generated Images? *Fencing Hallucination* explores the answers to these questions.



Figure 2: Chronophotograph *Fencer*, shot by Georges Demeny, Etienne-Jules Marey’s assistant, in 1906.

Chronophotography, invented by Etienne-Jules Marey in 1880s, was used to study human motion and later inspired many forms of visual art.[Braun 1992] Fencing was one of the sports analyzed by chronophotography. The unique look of chronophotographs[fig. 2], which involves overall motion pattern and complicated detail, is challenging to synthesize for even the best AI Image generator today. Taking on Lev Manovich’s proposal of AI Aesthetics: to create something between “art for computers” and “the universe of all aesthetic possibilities already realized in human civilizations,”[Manovich 2019] *Fencing Hallucination* aims to generate the human-understandable form of an image, a chronophotograph, with the latest AI tools. The generation process includes utilizing the audience’s locomotion data to directly guide the AI image generators, resulting in images that involve more human participation and feel more personalized to the audience. In the form of a Fencing game with the virtual AI Fencer on the screen, the human-AI interaction creates a chronophotograph unique to each audience’s experience.

2 RELATED WORK

With its continuing research advancements in the past decade, Deep Learning, generally interpreted as AI, has increasingly been used in creative AI tools. The most common creative AI is AI Image Generators, such as Style Transfer[Gatys et al. 2016], StyleGAN2[Karras et al. 2020], and most recently, diffusion models[Rombach et al. 2022; Song and Ermon 2019]. Diffusion models, such as *DALL-E 2*, are famous for their ability to use natural language to guide the generation or editing of the produced image, on top of the State-of-the-Art fidelity and varieties in their result images. Although potentially supported by diffusion models, other modalities such as semantic map, edge map, and keypoints for controlling the result appearance need more exploration.[Zhan et al. 2021] *Fencing Hallucination* explores the possibility of using human locomotion as an extended modality to control the creative AI tool.

Locomotion has been a longstanding theme in art. Since the 1880s, *chronophotography* was invented to record the sequence

of human motion in one photo to study the phase of the movement.[Braun 1992] It profoundly influenced later art movements such as Cubism, Futurism and the history of cinematography. Locomotion nowadays has been represented as numerical data and used in creative film and animations[Akten 2011], games[Ján Ilavský 2018], visualizations[Zhang et al. 2018], and interactive art[Vincent Houze 2018]. However, creating locomotion-based interactive art with AI tools is challenging because many AI tools, including diffusion models, cannot run in real-time due to their high computational resource demands. There are few real-time interactive AI artwork[Memo Akten 2017; Meshi and Forbes 2020; Sihwa Park 2020; Zhang and Ren 2021] that could benefit from higher generation qualities and greater flexibility.

Fencing Hallucination mitigates the conflict between interactivity, modality variety, and computational limits in using AI tools to create artwork. It provides a real-time human-AI interaction experience in the form of a fencing game. It also extends the modality to control the diffusion model by using locomotion as the input. Through a multi-step method, it generates a series of human and AI fencing images and then additively blended them into a chronophotograph image. The final result preserves the intricate details in each photo in the series and reveals the overall motion pattern.

3 FENCING HALLUCINATION INTERACTIVE AI INSTALLATION

The experience of *Fencing Hallucination* was implemented with two pipelines: the realtime interactive pipeline and the offline non-interactive pipeline. [fig. 3] The audience movement influences the AI fencer’s pose, triggers the strobe light, and guides the generation of chronophotographic outcomes.

3.1 Real-time Virtual AI Fencer

Captured by a Kinect motion sensing device as skeletal data of a matrix of $N \times 3$ (N is the number of joints), the visitor’s pose at a given moment is fed into a Multilayer Perceptron(MLP) to generate the AI fencer’s skeletal data. The MLP was also configured with a 2D distance parameter to control how far the generated AI fencer is from the audience’s skeleton in the screen space. Lightweight enough, the MLP runs in real-time to create the pose of the virtual AI Fencer when the audience moves. Please refer to the real-time interactive part in [fig. 3]. To enhance the experience, AI fencers are generated at different distances and shown sequentially to give an illusion of the AI fencer approaching the audience and performing a lunge. The screen show the real-time rendering of the two skeletons using TouchDesigner. This real-time interactive module provides an interface for the audience to input data into the following diffusion model.

3.2 Chronophotograph Synthesis

The original chronophotograph consists of multiple exposures of a subject in motion on a single sheet of film. The exposure process is simulated by blending a set of images additively. The simulation of *chronophotography* in *Fencing Hallucination* uses a multi-step pipeline.

First, TouchDesigner produces screenshots at different moments during the visitor’s interaction. The screenshots do not contain the

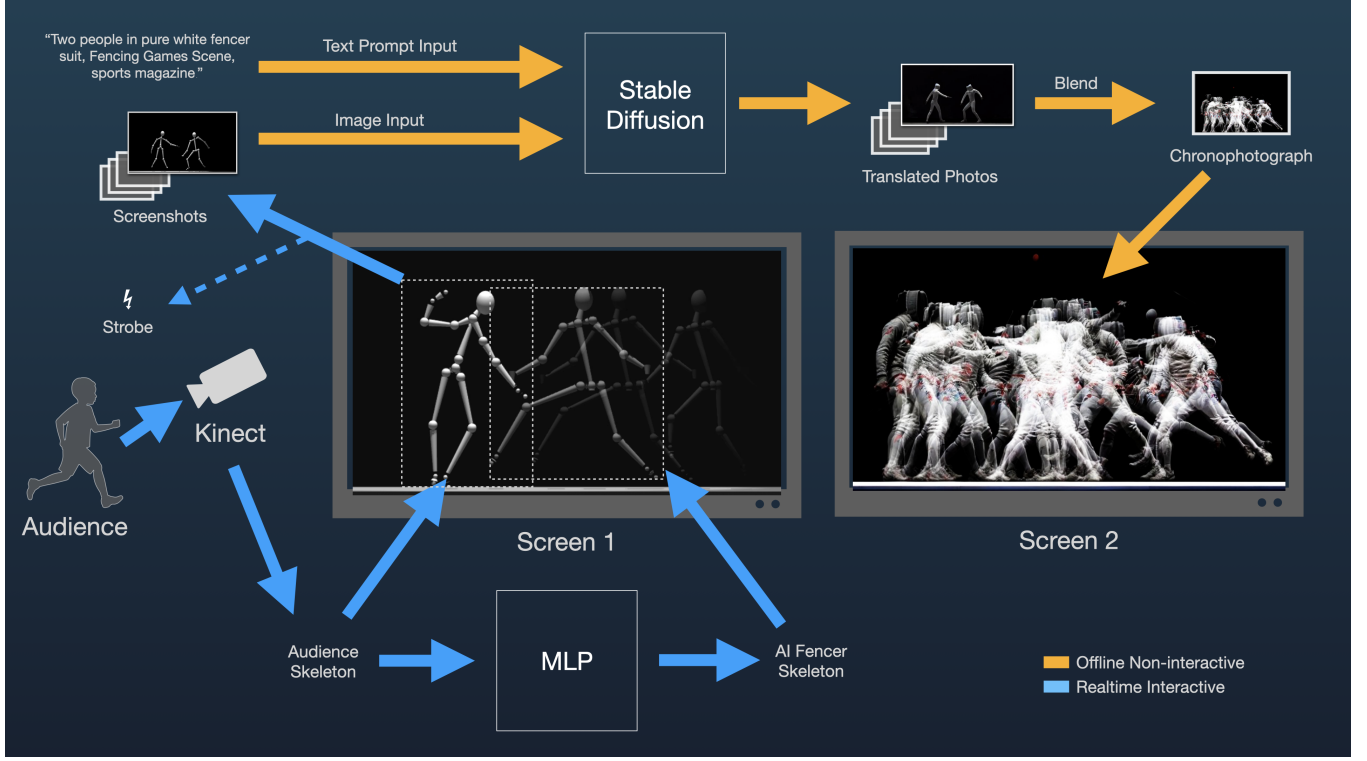


Figure 3: System Illustration. The system consists of two data flows: the real-time interactive (labeled in blue) and the offline non-interactive (labeled in orange). It starts with the audience captured by Kinect. A MLP is used to generate the AI Fencer’s skeletal data. The skeletons of both the audience and the AI Fencer are presented on the screen as stick figures. The audience’s motions trigger the strobe light when it reaches a level of activeness. In the background, the stick figure screenshots are translated into realistic photos by a stable diffusion model, and then are finally blended into the chronophotograph.

approaching illusion, showing only the two skeletons of the audience and the AI fencer. The diffusion model performs the image-to-image translations on each screenshot. With a customized prompt and a fixed seed value, the diffusion model synthesizes a series of images with a consistent style. These AI-synthesized images are blended in "additive" mode to produce the final chronophotograph-styled image. The blending equation is defined as: $c = 1 - (1 - a) * (1 - b)$ in which a and b is the pixel being blended, and c is the result after blending two pixels. Please refer to the offline non-interactive part in [fig. 3]

The selection of images to blend into the final chronophotograph has an important influence on the final appearance. We have experimented with two methods. The first is to select images at an equal time interval of I frames. The second is to filter out the images captured when the user is highly active, with T as the threshold of the user’s activeness value. The results from different I and T parameters can be seen in [fig. 4]. The second method is chosen in the actual installation with $T = 0.21$.

3.3 Installation Design

Besides the two screens, the installation also includes a strobe flash to indicate the activeness of the visitor’s movement.[fig.5] The strobe flash is triggered when the activeness value exceeds a certain

threshold. It is implemented by connecting the strobe’s remote trigger to an Arduino-controlled digital circuit with a PC-sync cable. When the activeness value surpasses the selected threshold, it sends a Serial port message to the Arduino. Upon the Arduino receiving the message, it changes the state of the circuit to short the two ends of the PC-sync cable, which signals the remote trigger to flash the strobe.

The audience can only move in a confined Fencing area during the experience. It is designed so for two reasons. First, it follows the rules in the actual fencing game, giving the player a finite strip-shaped room to move. Second, it is more pragmatic to keep the distance from the audience to the Kinect roughly constant so the scale of the figure on the screen remains unchanged. The ideal setup accommodates a fencing area about 3 meters long when setting the Fencing area 2.5 meters away from the Kinect.

4 EXPERIENCE

The experience of *Fencing Hallucination* starts with an audience stepping into the designated fencing area and moving its body to interact with a virtual AI Fencer on the first screen. When the audience’s movement is rapid and active, it produces a more exciting response from the AI fencer and triggers a short period of high-frequency strobe flash. The second screen starts with a blank screen

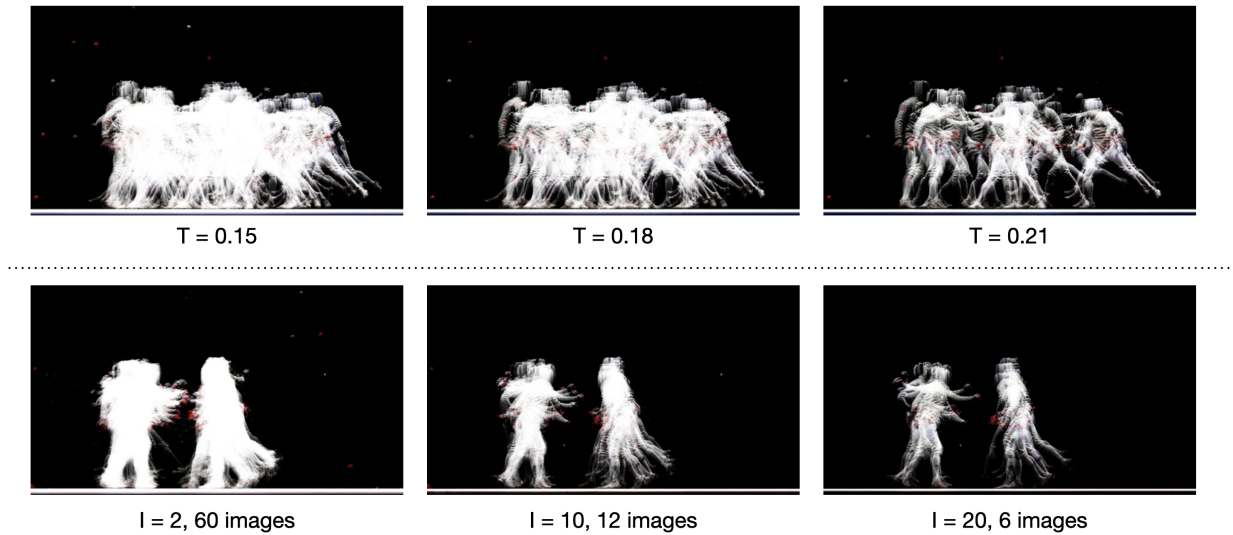


Figure 4: Different results with different threshold activeness(top row) or sample intervals(bottom row). The images in the top row were blended from samples of an 30-second experience, whereas the images in the bottom row from a 2-second segment in the 30-second period.

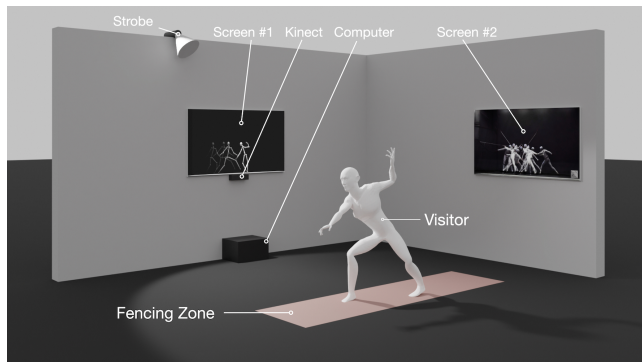


Figure 5: Installation layout.

and continuously refreshing itself to the latest chronophotograph generated from the human-AI interaction since the beginning. The image on the second screen becomes increasingly complex as the audience extends their interaction time. At the end of the experience, the audience can download the final chronophotograph by scanning the QR code at the corner of the screen. The recommended interaction time for each audience is between 30 seconds to 60 seconds.

When the installation is idle, the two screens show previous interactions and results to provide the potential audience with an experience preview. The first screen shows a pre-recorded Human-AI interaction. The second screen loops through a gallery of the

chronophotographs generated from the previous audiences' interactions. Once the audience steps into the fencing zone, the screens switch to the content stated in the above paragraph.

5 CONCLUSION

Fencing Hallucination explored the interactivity between a human and an AI and creatively visualized the interaction with a chronophotograph. In synthesizing the chronophotograph, it also explored using body movement as another modality to guide diffusion models to generate images. In the future, we plan to generate the virtual AI fencer's pose with more advanced Neural Networks to achieve more natural response to the audience's movement.

REFERENCES

- Memo Akten. 2011. *Forms* (2011). (2011). <https://www.memo.tv/works/forms/>
- Marta Braun. 1992. *Picturing time: the work of Etienne-Jules Marey (1830-1904)*. University of Chicago Press.
- Leon Gatys, Alexander Ecker, and Matthias Bethge. 2016. A Neural Algorithm of Artistic Style. *Journal of Vision* 16, 12 (2016), 326–326.
- Peter Hrinčár, Ján Ilavský, and Vladimír Hrinčár. 2018. Beat Saber. <https://beatsaber.com>.
- Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2020. Analyzing and improving the image quality of stylegan. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8110–8119.
- Lev Manovich. 2019. Defining AI arts: Three proposals. *AI and dialog of cultures" exhibition catalog*. Saint-Petersburg: Hermitage Museum (2019).
- Memo Akten. 2017. Learning to See (2017). <https://www.memo.tv/works/learning-to-see/>.
- Avital Meshi and Angus G Forbes. 2020. Stepping inside the Classification Cube: An intimate interaction with an AI system. *Leonardo* 53, 4 (2020), 387–393.
- OpenAI. 2022. Dall-E 2. <https://openai.com/dall-e-2/>
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10684–10695.
- Sihwa Park. 2020. UncertainFacing. <https://sihwapark.com/Uncertain-Facing>.

- Yang Song and Stefano Ermon. 2019. Generative modeling by estimating gradients of the data distribution. *Advances in Neural Information Processing Systems* 32 (2019).
- Vincent Houze. 2018. fluid structure 360. <https://vincenthouze.com/portfolio/fluid-structure-360/>.
- Fangneng Zhan, Yingchen Yu, Rongliang Wu, Jiahui Zhang, and Shijian Lu. 2021. Multimodal image synthesis and editing: A survey. *arXiv preprint arXiv:2112.13592* (2021).
- Weidi Zhang and Donghao Ren. 2021. Cangjie's poetry. In *ACM SIGGRAPH 2021 Art Gallery*. 1–3.
- Xiuming Zhang, Tali Dekel, Tianfan Xue, Andrew Owens, Qiurui He, Jiajun Wu, Stefanie Mueller, and William T Freeman. 2018. Mosculp: Interactive visualization of shape and time. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology*. 275–285.

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